Chapter 10. Driving with tests

This chapter covers

- Understanding unit tests
- Dealing effectively with dependencies inside tests
- How to test asynchronous programs
- How to test behavior
- Test suites and how to automate your test suite

> I saw the best minds of my generation destroyed by madness.
>
> *From Howl by Allen Ginsberg*

You have to solve harder problems in CoffeeScript today than most people had to solve in JavaScript in the early days. Back then, JavaScript was used to enhance browser applications. Today you're writing entire applications in CoffeeScript. Writing more of the program means dealing with more of the problems. Writing tests for your programs can help with these problems and save you from losing your mind. Writing the tests *first*, before writing the programs, is a technique referred to as *test-driven development*.

If you've never done it before, test-driven development can feel uncomfortable, like having your pants on backwards. Be patient and give it some time, though; it might take a while to adjust. If you've done test-driven development before but not in CoffeeScript, you'll be pleasantly surprised at how easy some of the techniques are.

This chapter presents and encourages a test-driven approach to development and leads you through the creation of your own tiny test framework. In practice, it's more likely you'd use an existing test framework, but your understanding will be deeper if you go through the steps to build a tiny framework. The techniques for writing unit tests, dealing with dependencies, testing asynchronous callback-driven code, and structuring